PROJECT MANAGEMENT WITH A CONTRACTOR

By
DUDLEY J. KINGSNORTH[1]

ABSTRACT

The long term viability of a mining company is dependent on the success of its capital investments. Due to the specialised nature of the design, engineering and construction of large projects the management is often contracted to an outside organisation. This paper is a review of the phases of the relationship between Client and Contractor. A rationale is developed upon which the selection of a Contractor can be made. Subsequently, the key aspects of the Client/Contractor relationship are reviewed with reference to completing the project on time within budget. The aspects discussed include Contractor management systems, qualifications of key personnel, definition of scope of work, industrial relations, project control and Contractor remuneration.

INTRODUCTION

At any time a large organisation has a number of possible projects under consideration, all of which are competing for the same limited investment capital. The contenders for selection will include both new projects and additions to existing operations, each of which will have their own peculiar set of problems.

It is essential that once the viability of a project has been established, and it has been given the go-ahead by Management, that it be executed in a timely manner within budget. Very few mining companies are of sufficient size to undertake the design, engineering, procurement and construction management of a large project (>$50 million) in-house. Consequently a Principal Contractor (or Engineer/ Construction Manager, referred to as the Contractor) is usually appointed to fulfil this role under the direction of the owner or his nominee. (In the case of a joint venture this may be the management company. Hereafter the owner/developer/manager will be referred to as the Client). The success of the project is dependent upon the subsequent direction/management of his activities by the Client.

[1]Senior Project Engineer
Shell Co of Aust Ltd, Bunbury W.A. 6230

This paper is a review of the phases of this relationship, drawing attention to the means whereby the project can be completed within budget and on time.

SELECTION OF A CONTRACTOR

Contractors spend a great deal of time and effort in the preparation and submission of proposals to a client. To ensure that only fully interested and qualified contractors are asked to submit proposals, a short list should be prepared upon the basis of responses to an invitation to prequalify. This should reduce the number of prospective contractors to a manageable four or five (each of whom has a reasonable chance of success, thereby ensuring a good submission on their part).

The most appropriate method of selecting a Contractor, assuming that bids/proposals have been submitted by a number of organisations, is one based upon a ranking and rating evaluation (i.e. a matrix-type evaluation). For the selection to be objective it is desirable that the ranking and rating be established prior to invitations to bid being issued, so that the proposals are submitted in a form that assists the evaluation. A typical evaluation is shown in Table I in which a number of key criteria are identified.

Contractor Management System

A large project typically involves the award of over fifty major sub-contractors (>$1 million) and the placing of several hundred purchase orders, any of which could be critical to the viability of the project. The effective management of the scheduling, preparation, evaluation of proposals, awards, supply and completion of these purchase orders and subcontracts is essential to the success of the project.

Evaluation of proposals from a Contractor should include an in-depth review of the proposed management systems. This should take the form of an explanation by the Contractor

The Aus.I.M.M. Sydney Branch, Project Development Symposium, November 1983

TABLE I - EVALUATION OF CONTRACTORS FOR A PROJECT

Criteria	Key Questions/Aspects	Rating	*Ranking	Max. Points
Quality of Management Systems.	Cost Control. Construction Management. Scheduling. Procurement. Engineering Capabilities. Inspection & Expediting. Is the control system computerised?	25		100
Qualification of Key Personnel.	Industrial Experience. Local/Australian Experience. Have they worked as a team before? Availability.	20		80
Corporate Qualifications.	What is past record? Strengths and Weaknesses of Organisation. Other Commitments. Image with Local Sub-contractors. Corporate Back-up (e.g. specialists available).	20		80
Industrial Relations.	Past Record. What is the philosophy to be adopted for this project?	20		80
Reimbursable Costs.	Basis. Overheads, Burden, etc. Local Office Costs. Site Costs. Method of Charging Consultants. Calculation of Fee.	16		64
Project Structure, Type of Organisation.	How does it 'fit' with Client Accountability/Authority of Individuals. Is it a Joint Venture? Location of Design Office. Project Procedures.	15		60
Quality Control.	Quality Control Procedures. Overseas Arrangements.	12		48
Proposed Project Schedule.	Is it realistic? What is the basis? Expediting Requirements.	10		40
Realism shown in Proposal.	Number of Manhours, Drawings, etc. required. Are the assumptions of the information to be supplied by the Client reasonable?	8		32
Access to Records.	Consider Drawings, Bids, etc. Calculations to be handed over on Completion of Project. Filing/Distribution System.	6		24
Contractor Liability.	(Note: In the case of turn-key project, as opposed to cost plus, this would be accorded a higher ranking).	4		16
	TOTAL			624

*Ranking is on a scale of 1 to 4
4 = excellent
3 = good
2 = acceptable
1 = poor
0 = unacceptable.

of a typical Monthly Report that he would submit and how this document would be used to control the project. Deviations from the budget/plan and the corrective actions taken should be highlighted without resorting to excessive detail. The presentation on the management systems should be made by the potential Contractors' nominated project team to demonstrate that it is workable and well understood.

A significant amount of the recording associated with the execution of a project is routine. It forms the basis for progress payments and the revision of schedules. To assist in the efficient recording of these details and to enable the information to be available immediately, the system should be computerised. The Client should not have to fund the development of a system for his project, as it should be included in the overhead costs. (Alder, 1983).

Qualification of Key Personnel

In many cases the final selection will be based upon the best "team" that is offered by the Contractor to undertake the engineering, procurement and construction management. The senior personnel in the project management team should have some experience in the relevant industry. The team that is offered must be balanced with no undue influence of engineering, procurement or on-site construction management such that decision-making is biased or distorted. Those personnel with process responsibilities should have some operating experience in addition to design/engineering expertise. In the case of an international consultant, Australian experience by a majority of senior personnel is considered highly desirable. A working knowledge of Australian industrial relations, relevant Standards and the local statutory regulations will assist in the expeditious completion of the project.

Proposals submitted by a joint venture of contractors or of consultants have to be considered carefully. Whereas collectively they may have the experts in a number of fields, often they have no proven record as a team. When the demands of a particular project are best met by the appointment of a joint venture, it is preferable that one of the joint venturers has an overall responsibility to simplify the development of procedures.

A significant number of large projects are managed by international organisations who can draw upon a workforce that is scattered throughout the world. It is imperative that the people who are selected to form the project team can work well together. In the case of key specialists (who's input will be required throughout the project), it is advisable to seek confirmation that the nominated individuals will be totally available to the project during the design and construction. The worst situation is where a specialist has a "part-time" role from afar on several projects.

Corporate Qualifications

The Contractor's corporate organisation should have sufficient depth of experience and expertise to provide the necessary back-up should the unexpected occur, e.g. exceptionally poor soil conditions, a lack of skilled people locally, or unforeseen design problems. To resolve such problems will probably only require the services of specialists for a limited time at short notice, but they have to be available if the project is not to be delayed with all the attendant cost implications.

An established "track record" in the construction of a given type of plant (e.g. petroleum refinery, long distance pipeline, or copper smelter) will assist in the rapid preparation of specifications and drawings. In its better form it will entail the Contractor to simply pull tried and tested designs from a drawer, with revisions for recent technological advances where appropriate. "Educating" a Contractor can prove a costly and frustrating exercise. Relevant corporate experience on the part of the Contractor will ensure realism in his estimating and scheduling.

The Contractor must be committed to 'your' interests in the project. If he is undertaking a very large number of projects then the assignment of his all too few 'top people' may be rationed. Conversely, if it is the only project the organisation is managing then there is a strong temptation to justify that an excessive number of people are assigned to the venture.

Consider the reputation of the Contractor. He should have a reputation as being fair and responsible with prospective sub-contractors. A Contractor with excessive and unnecessary contractual procedures can cause sub-contractors to add 10-15% to their quotations for the associated delays.

In summary, the strengths and weaknesses of the Contractor should not preclude a good working relationship with the Client organisation.

Industrial Relations Management

The Client must develop an overall

strategy for the management of industrial relations on site prior to the appointment of a Contractor.

The Contractor's proposal should include a detailed plan of how the industrial relations will be managed on site in accordance with the Client's strategy. Only by adopting a clearly defined plan from the outset can a Client retain an 'arms length' relationship with the Contractor on industrial relations, which is essential if there is to be no "flow-on" of construction conditions (such as living away from home allowance), to the operational phase.

The Contractor's past record in the management of similar projects, and this record's comparison with national trends and averages should be considered.

The experience and reputation of the Contractor's proposed industrial relations officers with the local union officials are of prime importance in the evaluation of this aspect of the proposal.

Contractor Remuneration

The cost of providing the engineering, procurement and construction management services will typically be in the range of 8 to 12% of the total project cost. In many cases it will be the largest contract of the project. The calculation of the cost of these services must be defined unequivocally, and agreed prior to award.

Evaluation of the proposals received should be completed on an equivalent basis in the first instance as agreements with a Contractor usually take the form of a 'cost plus' contract. The cost should be calculated on the estimated number of direct man-hours required to complete the project. This is necessary to ensure that the Contractor who has the lowest estimate of the cost of the project in terms of total man-hours expended (broken down into the various disciplines), does not receive any undue advantage in the analysis. It is recognised that this may have to be adjusted to compensate for any particular expertise one organisation may possess, [e.g. computer aided drafting (CAD) is more efficient than conventional drafting, but attracts a higher unit rate.]

THE MANAGEMENT OF A CONTRACTOR

Once the selection has been made there is a great possibility for the Contractor to be instructed to mobilise instantly, which sets in place a significant migration of personnel from around the world to the nominated offices/site! This temptation should be resisted while the details of the contract with the Contractor are put in place and relationships between key personnel are developed such that the total project team of Client and Contractor personnel have a clear and common understanding of the project and how it is to be implemented. The key steps in this early formative stage are discussed below:

Definition of Scope of Work

During the bid and evaluation stage the amount of information given to prospective contractors is restricted. Assumptions have to be made by the Contractor with respect to the amount of information he will receive upon award. The first step upon commencement is a joint review of the information to be supplied by the Client. It will include such documents as the project parameters, feasibility studies, environmental impact statements, soil data, flowsheets, design criteria, etc.

As a result of this study it may prove necessary to revise the schedule to allow some preliminary studies to be completed prior to full mobilization of the project team. An evaluation of several key alternatives, such as the transportation of the raw material, may be required before the scope can be defined in sufficient depth for detailed design engineering to begin. Such studies would be undertaken by small teams of specialists, often seconded from the Contractor's corporate office. Premature mobilization of the design team can lead to "wheel spinning", and a feeling by engineering personnel that the project lacks direction with the attendant losses of motivation and momentum.

The objective of these early discussions is to develop a sufficiently detailed scope of work with associated flow-sheets and basic design criteria for detailed engineering to commence. On a large project, such information may be released in 'packages' as alternatives for other areas are evaluated in parallel.

Items requiring a long lead time for procurement and construction such as power generating facilites, large rotating machinery and high technology equipment that has to be imported should be finalised first. Preliminary material take-offs will also be required so that orders may be placed for "bulk" steel.

Project Control

As part of the early feasibility studies a budget capital cost will have been prepared upon which the viability of the project will

have been proven, (with a probable accuracy of ±20%).

The proposed key project control documents should be reviewed and the basic format of the monthly/periodical report established. It is recommended that a 'Uniform Estimating Guide' be prepared jointly by the Client and Contractor. This will ensure that the bases for estimates/budgets prepared by the Contractor are similar to those prepared by the Client organisation, thereby obviating the need for any translation of the figures.

Once the project scope has been finalised the Contractor should be instructed to prepare a preliminary estimate. On a typical project of $500 million requiring a 3 year construction period this could take 6-9 months as preliminary quotations on the major equipment will be required for its preparation. If the selected Contractor has a good bank of data from a large number of similar projects this time could be significantly reduced. An accuracy of -10% to +15% is typical for such an estimate. Upon completion of this document, which breaks down the costs by major facility and character of work, it will act as a control in the management of costs.

A definitive estimate with a probable accuracy of ±7.5% will be prepared when engineering is 60-70% complete. Submission of these documents to the Client's senior management enables them to monitor the project continually.

In parallel to the above activities a detailed project schedule, utilizing computerized CPM techniques, should be prepared. The results of the analysis will also have an impact upon costs and consequently there should be regular meetings between both Client and Contractor's costing and scheduling personnel at which key decisions are documented for future reference.

For the project to be completed on time within budget it is essential that the Client fulfils his responsibility to meet the Contractor's performance demands. The appointment of a Contractor does not relieve a Client of all decision making. Tough and unpalatable decisions often have to be made rapidly if a tight schedule is to be maintained.

Project Procedures

The large contracting organisations have well developed project procedure and control systems with which their personnel are familiar. As the existence of such an internally well understood and effective system is a prime reason for selecting the given organisation its basic form should not be changed by the Client. The Contractor has an obligation to "educate" the Client in the use and interpretation of his systems. The Client should review the sytem in detail with the Contractor, and then clearly define the information that he wishes to receive on a regular basis. It is not necessary for a Client to review every document produced by a Contractor, provided he has access to all documentation when necessary.

On a large project both Client and Contractor organisations will nominate a project manager through whom all official contact and decision making will be made. Whilst such an arrangement may be ideal from both theoretical and legal viewpoints, practically, it is inefficient and the cause of many delays. It is better to establish a matrix-type of relationship with Client/Contractor contact at several levels each of which has an attendant authority level.

In establishing the project procedures optimum use should be made of current communication tools - e.g. computer to computer links - which assist and speed-up communications between the different disciplines.

One aspect of the project procedures that is often glossed over is the Document Distribution Schedule. It is worth spending some time evaluating the real need for the distribution of key documents as a means of developing an efficient organisation. Where possible, use should be made of summaries and variance reports. Every piece of paper that crosses the supervisor/manager's desk should provide him with information that will assist him in his function. Information distributed on a "nice to know" basis is worthless and time-consuming. For such information, a reading file should be established, which gives it a lower priority. A matrix-type organisation is consistent with this policy. (Estes and Ruskin, 1980).

Manage Design Changes

The most important factor contributing to budget overruns and project delays are continuing design changes. Engineers often fail to recognise the law of diminishing returns, in striving to achieve the perfect design. At some stage the design has to be "frozen", subsequently, the only justification for revision will be that the proposed design is either unsafe or unworkable. Ideally, a detailed feasibility study would enable the design to be frozen from the start, or failing

that, when the major alternatives have been evaluated as detailed above.

If the design is to be frozen, it must be accepted that some plant modifications will be required during the commissioning stage to bring the project up to design capacity. This expenditure would be included in the contingency allowance.

Industrial Relations

As soon as the Contractor has been appointed, that is prior to any mobilisation on site, the details of the industrial relations strategy should be developed. Matters such as the implementation of a site award versus use of existing federal/state awards, the appointment of industrial officers on site and their role, and site allowances should all be discussed.

To prevent the possibility and flow-on of undesirable practices to the operational phase, it is preferable for the Client to retain an arms-length relationship with the Contractor on industrial relations matters. This may be facilitated by the appointment of an employer organisation to co-ordinate industrial relations management on site.

Good industrial relations are fundamental to the timely completion of a project, a lost time figure of 15-20% can delay a project with a 3 year schedule by six months or more. Not only are there the direct time losses but once a site has gained a reputation as "industrially active", productivity will drop through lack of continuity and the better tradesmen will often leave for other projects where their income can be guaranteed.

The industrial relations representative on site may be likened to a doctor on a safari. If there is no "sickness" then the sceptics will re-state their belief that such people are not required, while ignoring any beneficial effect that preventative measures have had. However, if there is a dispute he has to pull the right panacea from his bag instantly. In selecting an industrial relations officer do not consider the witch doctor who manages by fear, or the brain surgeon who will spread confusion through his unnecessarily technical language. The incumbent should be likened to a G.P. who can communicate at the grass roots level, and will listen with patience to the many trivial complaints before offering sound, acceptable advice. In times of crisis he should be able to seek the advice and assistance of a specialist in the corporate organisation, (e.g. in the event that an appearance before an Industrial Commissioner is necessary or there is a need for direct negotiation with State or Federal Union officials) when qualified and knowledgeable support is required. In drawing this analogy, it is not an attempt to trivialise this aspect of project management, but rather to emphasise it.

Inspection and Expediting

Often little importance is given to inspection and expediting by the Client, and once the project is underway, assignments are left to the discretion of the Contractor. As both inspection and expediting are a form of insurance, and the cost of assignments are fully reimbursable the unsupervised Contractor will tend to be cautious. In some cases excessive expediting/inspection can sour future relations between the supplier and Client. Therefore, the Client should assign the responsibilities of the co-ordination and approval of expediting and inspection to specific functions in his organisation to ensure consistency in their application. Obviously, the two responsibilities should not be assigned to the same office, due to potential conflict of interest.

The key inspection and test reports must be retained as the basis for the maintenance information system during operation. The transfer of this information should be considered early in the project.

A Client representative should always be present at discussions/negotiations with statutory and licensing authorities to maintain continuity through to the operational phase.

Office Location

The majority of resource developments in Australia are in remote locations, while the design/procurement office has to be established in a covenient major city. The choice of the office location should take into consideration such factors as communications with the site, the Client's parent organisation, statutory authorities and the vendors and subcontractors.

It is considered preferable that the Client be allocated space in the same location as the Contractor for the following reasons

- the Client personnel can concentrate their effort on the successful completion of the project,
- the Contractor has ready access to the Client for decisions, and
- it assists in the rapid establishment of good rapport and communication

between Client and Contractor,

The timing of the move of key personnel to site is a matter worthy of some consideration. This move should be made as soon as design is complete, to leave a small representation in the original design/procurement office.. Once design is complete the major problems/decisions relate to activities on site and it is therefore logical for key personnel to be located on site.

CONCLUSION

A project can be completed on time and within budget if it is factually conceived, realistically and knowledgeably budgeted and scheduled, and decisively designed and constructed. This paper has identified the key attributes of a successful project management relationship between Client and Contractor.

ACKNOWLEDGEMENTS

The author acknowledges the informal contributions of his colleagues in Shell and the engineering/construction industry to this paper.

The author wishes to thank The Shell Company of Australia Limited for permission to present this paper. However, he wishes to record that he is responsible for the views expressed herein.

REFERENCES

J. ALDER (1983) - "Use of computerised management systems for the Blair Athol Coal Mine Project" - AusIMM Symposium on Computers in Mining, Brisbane, May 1983.

W.E. ESTES and A.H. RUSKIN - "The Organisation: Its Effect on Project Managers", American Society of Civil Engineers Convention and Exposition, Florida USA, October 1980.